
Your Signature _____**Instructions:**

1. For writing your answers use both sides of the paper in the answer booklet.
2. Additional sheets taken, if any, should be properly attached to the main answer booklet.
3. Please write your name on every page of this booklet and every additional sheet taken.
4. If you are using any result from class then please verify that the required hypothesis are satisfied.
5. Maximum Score: 100

Score

Q.No.	Alloted Score	Score
1.	20	
2.	25	
3.	30	
4.	20	
5.	25	
Total	120	

Number of Extra sheets attached to the answer script: _____

Time Allotted is 3 hours. Exam ends at 5pm.

1. Meteorologist Chakrapani could not predict rainy days very well in the wet city of Rainpuram. So he decided to use the following prediction model for rain. If it had rained yesterday and today, then it will rain tomorrow with probability 0.5. If it rained today but not yesterday, then it will rain tomorrow with probability 0.3. If it did not rain today, then it will rain tomorrow with probability 0.1. regardless of yesterday's weather. Let $X_n = R$ if it rained on day $n \in \mathbb{N}$ and $X_n = D$ if it was a dry day (no rain). Suppose $X_0 = D$ (so it didn't rain on day 0).
- (a) **(10 points)** Show that $\{X_n : n \geq 0\}$ is not a Markov chain.
- (b) **(10 points)** For the Markov chain $Y_n = (X_n, X_{n+1})$ write down the state space and the transition matrix for the chain Y_n .

2. Let $L \in \mathbb{N}$, X_n be a Markov chain on $S = \{0, 1, \dots, L\}$ with transition matrix $\mathbf{P}$ with

$$p_{ij} = \begin{cases} 1 & \text{if } i = j \in \{0, L\} \\ \frac{1}{2} & \text{if } i \geq 1, j = i + 1, i \in \{1, \dots, L - 1\} \\ \frac{1}{2} & \text{if } i \geq 1, j = i - 1, i \in \{1, \dots, L - 1\} \\ 0 & \text{otherwise} \end{cases}$$

- (a) **(5 points)** Sketch the graph induced by this Markov chain on the vertex set S when $L = 4$.
- (b) **(5 points)** Decide whether the chain is irreducible and determine its periodicity.
- (c) **(5 points)** Find the stationary distribution π of the chain.
- (d) **(10 points)** Let $T^{\{a\}}$ be the hitting time of $a \in S$. Let $h : S \rightarrow [0, 1]$ be given by

$$h(i) = \mathbb{P}(T^{\{0\}} < T^{\{L\}} \mid X_0 = i).$$

Find h .

3. Assume that there is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, on which $\{X_i\}_{i \geq 1}$ are all i.i.d. X and let $S_n = \sum_{i=1}^n X_i$ for $n \geq 1$. Let $\mathcal{A}_n$ be the observable events with regard to $\{X_i\}_{1 \leq i \leq n}$. Suppose $\mathbb{P}(X = 1) = \frac{1}{2} = \mathbb{P}(X = -1)$.

- (a) **(5 points)** Let $J = \min\{n \geq 1 : S_n = -a\}$ for $a \in \mathbb{N} \cup \{0\}$. Show that J is a stopping time with respect to $\mathcal{A}_n$ —being observable events with regard to $\{X_i\}_{1 \leq i \leq n}$.
- (b) For $\theta \in \mathbb{R}, n \geq 1$, let

$$Z_n = \exp(\theta S_n) (\cosh(\theta))^{-n},$$

with $\cosh(\theta) = \frac{1}{2} (\exp(\theta) + \exp(-\theta))$

- (c) **(10 points)** Show that $\{Z_n\}_{n \geq 1}$ is a martingale.
- (d) **(5 points)** Let J be as in part(a). Show that, when $\theta < 0$, the $\{Z_{\min\{n, J\}}\}_{n \geq 1}$ is a sequence of bounded random variables.
- (e) **(10 points)** Let J be as in part(a). Show that¹ for $0 < \alpha < 1$, one has

$$\mathbb{E}[\alpha^J] = \left(\frac{1 - \sqrt{1 - \alpha^2}}{\alpha} \right)^a.$$

¹You may use the Bounded convergence theorem: namely if $\{X_n\}_{n \geq 1}$ are a sequence of bounded random variables and $X_n \rightarrow X$ with probability 1 then $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$

4. Assume that there is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, on which $\{X_i\}_{i \geq 1}$ are all i.i.d. X and let $S_n = \sum_{i=1}^n X_i$ for $n \geq 1$. Let $m = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}[X]$.

(a) **(10 points)** Assume $m = 0$. Find a sequence of real numbers b_n so that $Y_n = S_n^2 - b_n$ is a martingale.

(b) **(10 points)** Show that for any positive integer n and $\epsilon > 0$ that

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k - km| \geq n\epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2}$$

5. At time 0, an urn contains 1 black ball and 1 white ball. At each time $n \geq 1$, a ball is chosen from the urn and returned to the urn. At the same time, a new ball of the same colour as the chosen ball is added to the urn. Just after time n , there are $n + 2$ balls in the urn of which $B_n + 1$ are black. Let M_n be the proportion of balls in the urn that are black, at time n .
- (a) **(5 points)** Show that M_n is a martingale.
 - (b) **(10 points)** Let T be the number of balls drawn until the first black ball appears. Show that T is a stopping time and find $\mathbb{E}[\frac{1}{T+2}]$.
 - (c) **(10 points)** Show that M_n converges to a random variable M as $n \rightarrow \infty$ with probability one and find the distribution of M .

November 12th, 2025

Introduction to Stochastic Processes

Name (Please Print) _____

Final Exam

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